

SELECTIVE FEATURES AND BEHAVIORAL CAUSALITY UNDER VISUAL OCCLUSION IN A VLA POLICY

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ABSTRACT

A feature that reliably responds to a visual condition need not mediate its effect on behavior. We examine this distinction in a LIBERO-finetuned vision-language-action policy using target-linked visual occlusion, sparse transcoders of 18 language-prefix MLPs, held-out demonstration frames, and closed-loop interventions. On LIBERO-Spatial, 452 of 73,728 features satisfy an occlusion-selectivity rule; 83.72% of a capped 129-feature subset reproduce that selectivity on held-out frames. Yet injecting this subset closes only 0.16% of the normalized action gap. A discovery-ranked set of 100 attribution features closes 1.49% [0.65, 2.54], compared with 56.85% for all transcoder features and 73.26% for full-MLP patches. Restoration is asymmetric: the attribution set closes 12.52% [3.07, 22.29] of the occluded-to-clean action gap. This action-level effect does not yield reliable behavioral recovery: across five tasks and 60 episodes per condition, clean and covered success are 58/60 and 10/60, attribution restoration yields 8/60, and full-MLP restoration yields 59/60. A second suite reproduces small sparse-set injection effects. These results distinguish reproducible selectivity, finite action interventions and executed success, without establishing that no compact causal circuit exists.

1 INTRODUCTION

Vision-language-action (VLA) policies turn multimodal observations and instructions into robot actions. Their internal features offer potential explanations of why visual changes alter control, but three questions must be separated. Does a feature respond consistently to a condition? Does changing that feature alter the action? Does the resulting intervention improve task performance? A positive answer to the first question does not imply positive answers to the other two.

We study this separation for target-linked visual occlusion in a LIBERO-finetuned $\pi_{0.5}$ policy (Physical Intelligence et al., 2025; Liu et al., 2023). The target is the object identified by the task instruction. We construct rendered box and screen occluders, alongside a pixel-paint control, and train sparse transcoders to approximate language-prefix MLP computations. Feature sets are selected on discovery demonstrations and patched on disjoint test demonstrations. We compare selectivity-selected and gradient-attribution-selected features, include random and full-MLP controls, and test restoration during execution.

The central result is a dissociation rather than a recovered circuit. Selective features replicate their activation criterion, but the capped selective set produces little action-gap closure. Attribution identifies a small positive injection effect and a larger restoration effect. Neither tested small set reliably restores task success under the box occluder, while full-MLP restoration nearly recovers the clean baseline. A second suite reproduces the small injection effects; it does not provide a second-policy or closed-loop replication.

Our contributions are: (1) an empirical separation of selectivity, held-out action effects and behavioral recovery under controlled visual perturbations; (2) quantitative comparisons of feature selection, intervention direction, budget and token scope; and (3) explicit accounting for realized task coverage, reconstruction fidelity and reproducibility limits. The results constrain the tested dictionary and intervention procedures. They do not prove inherent distributedness or establish an object-permanence mechanism.

2 RELATED WORK

Sparse representations and causal interventions. Transcoders approximate MLP input-output maps with sparse features (Dunefsky et al., 2024). Unlike reconstruction of a single activation space, this formulation supplies decoded feature contributions to an MLP output. We apply this established representation to a VLA prefix and evaluate finite interventions. Activation-patching conclusions depend on metrics, corruption choices and controls (Zhang & Nanda, 2024); we therefore report both action-gap closure and closed-loop success, with uncertainty at the task level.

VLA and visual-model interpretation. Sparse-autoencoder interpretation and steering have been studied in VLAs (Swann et al., 2026), including comparisons between feature types (Grant et al., 2026) and event-grounded feature learning with behavioral evaluation (Jin et al., 2026). Transcoders have also been applied to visual grounding in vision-language models (Damianos et al., 2026). Thus neither sparse VLA features nor behavioral interventions are new in isolation. Our specific contribution is the joint evaluation of occlusion selectivity, discovery-ranked interventions, held-out transfer and behavioral recovery. We make no general claim that related studies lack held-out or causal evaluation.

3 METHOD

3.1 POLICY, DATA AND INTERVENTION SITE

The completed runs use the LIBERO-finetuned $\pi_{0.5}$ policy on Spatial and Object suites (Physical Intelligence et al., 2025; Liu et al., 2023). The policy conditions an action-chunk sampler on camera images, language and robot state. The implementation hooks MLP inputs and outputs in the 18 PaliGemma language-prefix layers. These are *not* action-expert MLPs. Transcoders do not have denoising-timestep conditioning; attribution differentiates through the action sampler to the prefix perturbations.

Spatial discovery uses demonstration IDs 0 and 1; test uses 2–4; extra training uses 5–16. Object uses discovery IDs 0 and 1, test IDs 2 and 3, and extra training IDs 4–11. Probe frames target phases 0.30, 0.55 and 0.80 of the approach before the first target lift exceeding 0.01 m, with a three-step pre-lift margin. A local search within six steps enforces visibility and edit-placement constraints, including a target mask of at least 80 pixels.

The realized Spatial sample contains 46 discovery and 70 test frames across eight tasks, despite ten configured tasks. Tasks 4 and 6 supply no valid probes. Object supplies 60 discovery and 60 test frames across ten tasks. Extra training uses 419 Spatial and 299 Object frames. Dictionary training includes discovery data; test demonstrations are excluded. The dictionary’s random token holdout is not an independent episode holdout.

3.2 VISUAL CONDITIONS

BOX inserts a render-only opaque box around the target, visible to both cameras. SCREEN places a panel between the third-person camera and target, leaving wrist-camera information available. PAINT replaces target pixels with gray. Each family has a segmentation-checked miss control. Box and paint also have distractor controls; screen does not. Base, recolor and absent conditions complete eleven observation conditions. Absence is rendered by temporarily moving the target away and restoring simulator state.

The box and screen are recomputed from live state and have no collision dynamics. Their location reveals target location, and their extent can hide contextual or robot pixels. The experiments therefore manipulate visual evidence without testing physical obstacle interaction. They do not directly test memory for a persistently hidden object. Counterfactual clean renders used for restoration are privileged diagnostics.

3.3 TRANSCODERS AND FEATURE SELECTION

For each layer ℓ , a TopK transcoder with 4,096 features and 64 active features per token approximates the MLP output:

$$\hat{y}_\ell(x) = D_\ell f_\ell(x) + b_\ell. \quad (1)$$

Across 18 layers there are 73,728 feature coordinates. Training uses normalized MSE plus a dead-feature auxiliary loss, Adam at learning rate 0.002, batch size 4,096 and 8,000 steps, with a 12,000-step retry when needed. Token holdout is 5%. Additional settings appear in Appendix A.

To patch feature set S from donor d into recipient r , we add

$$\Delta y_{\ell,t} = \sum_{j \in S_\ell} D_{\ell,j} (f_{\ell,j,t}^d - f_{\ell,j,t}^r) \quad (2)$$

to the real recipient MLP output. This preserves the recipient reconstruction residual. We separately patch all transcoder features, the residual error only, and full MLP outputs. Their measured effects are nonlinear and are not an additive causal decomposition.

Selectivity. Discovery-time selection sums activation over tokens. A feature must increase on at least 90% of frames, rise by at least one typical firing magnitude, and meet specificity 0.5 relative to recolor, absent and miss responses. A target-specific variant also compares a distractor. Capped sets retain at most eight features per layer. We report capped and uncapped replication separately.

Attribution. For each discovery frame, decoded feature differences are scored against gradients of a directional action-projection objective. The injection contribution has the form

$$\alpha_{\ell,j} = \sum_t (f_{\ell,j,t}^o - f_{\ell,j,t}^b) \left\langle D_{\ell,j}, \frac{\partial m}{\partial y_{\ell,t}} \right\rangle. \quad (3)$$

Injection and restoration endpoint scores are averaged, then aggregated over discovery frames. The top 100 alive features are ranked in layers 0–13, selected by a discovery residual-swap criterion. Full-transcoder patches span all 18 layers; this support mismatch limits a pure feature-count interpretation of sparse-versus-full comparisons.

Controls. Primary random controls match per-layer counts and decoded per-token edit norms, with repeated draws averaged within frame. Zero random edits remain zero. Budget-curve random controls use raw edits. Cross-task comparisons reuse controls without fold-specific rematching and retain the overall dictionary, so they are not entirely task-isolated training replications.

3.4 METRICS AND STATISTICAL ANALYSIS

For recipient action a_s , target action a_t and patched action a_p , we measure normalized action-chunk gap closure

$$C = 100 \left(1 - \frac{\text{RMSE}(a_p, a_t)}{\text{RMSE}(a_s, a_t)} \right). \quad (4)$$

Negative values indicate increased distance. A separate projection metric measures movement along the start-to-target direction. Injection transfers occluded feature values into the clean run; restoration transfers clean values into the occluded run. Neither metric is a language-logit score. Rollouts execute postprocessed actions.

Causal tests require an action gap of at least 0.1, distinct from the 0.02 validity threshold. Retained Spatial box/screen/paint frame counts are 70/69/63; Object box/paint counts are 59/48. Confidence intervals use 10,000 whole-task bootstrap resamples of a frame-weighted mean. Paired one-sided sign-flip tests enumerate signs over task-mean differences. The stored protocol sets a ten-point practical margin for the primary attribution contrast; no external preregistration is available. Exploratory tests have no multiplicity correction.

4 RESULTS

4.1 INSTRUMENT FIDELITY AND SELECTIVE RESPONSES

Median held-out-token FVU is 0.204 for Spatial and 0.183 for Object; on held-out test frames it is 0.294 and 0.334. FVU is variance *unexplained*. The Spatial reconstruction criterion fails, and non-

Table 1: Held-out injection: mean percent action-gap closure [95% task-bootstrap CI]. Full controls are shown as means; complete intervals and all controls are in the supplementary tables. Spatial uses eight task clusters; Object uses ten.

Intervention	Spatial box	Spatial screen	Spatial paint	Object box
Attribution, 100	1.49	2.81	5.25	0.90
95% CI	[0.65,2.54]	[0.44,5.01]	[3.31,7.47]	[0.46,1.41]
Capped selective	0.16	0.45	-0.01	0.20
95% CI	[-0.11,0.50]	[0.00,1.17]	[-0.21,0.15]	[0.04,0.38]
All transcoder	56.85	55.10	59.02	47.70
All MLP	73.26	78.54	75.80	74.10
Test frames	70	69	63	59

blocking configuration permits continuation. The dictionary is therefore an imperfect instrument, not a complete account of the policy computation. The saved batching check tests row composition at a fixed batch size, not universal single-item/batched equivalence.

Spatial discovery identifies 452 box-selective features (417 target-specific), 731 screen-selective features and 24 paint-selective features. Object identifies 457 box-selective features (439 target-specific) and 12 paint-selective features (nine target-specific). For Spatial box, 108/129 capped features reproduce selectivity (83.72%), while the uncapped rate is 341/452 (75.44%). Object box rates are 75.19% capped and 64.11% uncapped. These rates validate the operational activation criterion; they do not by themselves identify causal mediators.

4.2 SMALL INJECTION EFFECTS AND ASYMMETRIC RESTORATION

Table 1 reports held-out injection effects. Spatial box attribution exceeds matched random by 1.45 points [0.64, 2.47], with positive task means on all eight tasks (one-sided sign-flip $p = 0.00390625$). This is detectable but falls below the ten-point practical margin. The capped selective contrast is 0.12 [-0.14, 0.45]. Even all 452 selective features close only 0.46% [-0.25, 1.13].

All-transcoder Spatial box injection closes 56.85% [49.40, 67.82], versus 73.26% [67.46, 80.90] for full MLP outputs. The error-only patch closes 6.88%. These positive controls show that the intervention site carries a substantial action effect, while preserving uncertainty about information outside the learned dictionary.

Restoration is substantially stronger than injection. Spatial box attribution restoration closes 12.52% [3.07, 22.29], exceeding random by 12.79 points [3.35, 22.49]. This contrast meets the configured ten-point margin with an interval above zero. Screen and paint restoration close 7.29% and 18.73%; Object box closes 7.12% [3.01, 12.41]. A blanket claim that sparse interventions have no causal effect would therefore be incorrect.

4.3 BUDGET, RANKING AND LOCALIZATION

Discovery-ranked Spatial box injection grows from 0.19% at ten features to 13.61% at 1,000 and 23.65% at 3,000 (Figure 1). A test-frame-specific attribution diagnostic reaches 16.7% at 100 features and 29.2% at 3,000. This diagnostic uses the test frame and is not a held-out ranking or an upper bound on all possible circuits. Its advantage is consistent with context-dependent ranking quality as an alternative to intrinsic distributedness.

Only eight of the top 100 attribution features overlap the 452 Spatial box-selective features. Their selectivity-rise/attribution Spearman correlation is approximately -0.0194; Object box has zero top-set overlap and correlation -0.0151. Thus the operational rankings dissociate despite reproducible selectivity.

Full-MLP patches restricted to occluder-overlapping tokens close 45.62% [38.53, 56.79], versus 9.89% [4.86, 15.31] for other tokens. Corresponding attribution-set effects are 1.13% and 0.25%. These token-scope effects are not additive. Across pooled Spatial box intervention observations, predicted and measured projection correlate at $r = 0.28$, slope 0.06; screen and paint correlations

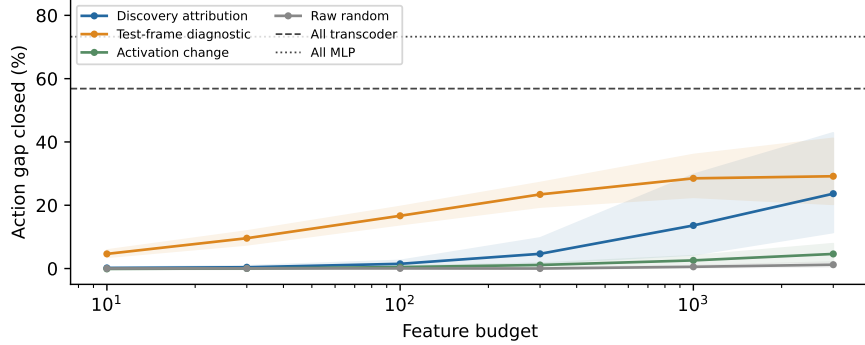


Figure 1: Spatial box injection versus feature budget. Bands: task-bootstrap 95% intervals. The test-frame diagnostic uses each test input for ranking and is not a held-out global method or optimal upper bound. Random budget controls use raw edits.

Table 2: Closed-loop success. Each condition contains 60 episodes across five tasks. Paired differences use task-bootstrap intervals; counts are not 60 independent task replications.

Condition	Successes	Rate (%)
Clean	58/60	96.67
Box	10/60	16.67
Screen	33/60	55.00
Paint	44/60	73.33
Box + restore all MLP	59/60	98.33
Box + restore attribution	8/60	13.33
Box + restore random	8/60	13.33
Box + restore selective	11/60	18.33
Clean + inject attribution	60/60	100.00
Paired contrast	Percentage points [95% CI]	
Attribution restore — box	−3.33 [−15.00, 5.00]	
Attribution — random restore	0.00 [−11.67, 13.33]	
Full-MLP restore — box	81.67 [68.33, 93.33]	

are approximately 0.74 and 0.77. Repeated frames and overlapping sets limit independent-sample interpretation.

Attribution’s advantage over random is 5.17 points on discovery frames and 1.45 on held-out frames. Cross-task attribution closure is 0.65%, with a reused-control contrast of approximately 0.62 points. The dictionary and control qualifications in Section 3 prevent treating this as fully isolated cross-task training.

4.4 EXECUTED BEHAVIOR AND SECOND-SUITE EVIDENCE

Closed-loop evaluation uses Spatial tasks 0, 2, 6, 7 and 9, twelve episodes per task, and nine conditions: 540 episodes total. The code requests matched initialization IDs and noise seeds; saved records do not contain hashes of actual reset states. Box is selected for restoration after its larger observed baseline drop relative to screen. At each inference step, restoration obtains a clean counterfactual render of the current state.

Table 2 shows a large visual-occlusion impairment and strong full-MLP recovery. Attribution restoration provides no reliable improvement over either covered or random-restoration baselines. Its normalized recovery fraction is -0.04 $[-0.24, 0.06]$, compared with 1.02 $[0.95, 1.16]$ for full-MLP restoration. Small action-level restoration therefore does not establish behavioral recovery. Injecting the attribution set into clean rollouts also does not measurably impair success in this sample.

Object-suite box injection reproduces the qualitative separation: attribution 0.90%, capped selectivity 0.20%, all-transcoder 47.70% and full-MLP 74.10%. There are no Object rollouts. An attempted second-policy run failed during gated tokenizer access before producing comparable results. Generalization is therefore across two suites for one policy, with behavioral evidence confined to Spatial.

5 DISCUSSION AND LIMITATIONS

The findings support a narrow but useful conclusion: reproducible activation selectivity does not identify the tested small feature sets as sufficient behavioral mediators. Attribution finds more action influence, especially in restoration, but does not repair execution. Full-MLP restoration is a strong positive control and a privileged intervention rather than a deployable occlusion-robust policy.

Several explanations remain unresolved. The sparse dictionary leaves reconstruction error, with a failed Spatial fidelity gate. Global rankings can miss context-specific features; finite patches need not follow first-order gradients; nonlinear interactions may make isolated contributions misleading. Ranked and full-feature interventions also differ in layer support. These possibilities prevent inferring that no compact circuit exists. A stronger identification study would compare layer-matched supports, context-conditioned rankings, alternative dictionaries and multiple seeds.

The occluders reveal target location and lack physics. The screen preserves wrist information. Demonstration probes emphasize pre-lift approach states; rollout distributions differ. Discovery overlaps dictionary training, and the second-suite results do not establish architecture-level generalization. There are only eight or ten causal task clusters and five rollout clusters; the many diagnostics are exploratory and uncorrected for multiplicity. Finally, missing replay frames, captured arrays and checkpoints prevent full inference-level reproduction from the uploaded package alone.

6 CONCLUSION

In the tested VLA, selective responses, action-gap mediation and success recovery are distinct empirical outcomes. Selective features reproduce their activation criterion, attribution yields measurable but limited held-out action effects, and tested small-set restoration does not reliably recover behavior. Reporting these stages together provides a more precise mechanistic claim than feature selectivity alone.

REPRODUCIBILITY STATEMENT

The accompanying audit inventories all 181 input files and verifies 14,370 intervention rows and 540 rollout episodes using the saved records. Appendix A and the full supplementary tables document configuration and results. This statistical verification does not rerun training or inference. The artifact manifest has 36 missing entries, including replay inputs and checkpoints; only seven Spatial layer checkpoints and no Object layer checkpoints are uploaded. These gaps limit complete reproduction.

ETHICS STATEMENT

The reported interventions are simulator-based diagnostics. No real-robot deployment or human-subject experiment is reported. Counterfactual access to clean observations is privileged, and the results do not justify safety guarantees for physical robots.

AI USE STATEMENT

AI assistance was used to inspect archived artifacts, prepare numerical verification code, organize the analysis, and draft and format this manuscript. No new policy training, simulator intervention or rollout was performed during manuscript preparation. Human authors must verify the scientific claims, references and final submission materials and retain responsibility for them.

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A CONFIGURATION AND COMPLETE-RESULT GUIDE

Training uses 396,551 Spatial and 386,859 Object tokens per layer, with holdouts of 20,871 and 20,361. The auxiliary TopK is 256, coefficient 0.03125, inactivity window 100, warmup 5% and decay 20%; seeds are based on zero plus layer index. Dictionary dimensions are 4,096 features and TopK 64 for every layer. The primary attribution budget is 100; injection budgets are 10, 30, 100, 300, 1,000 and 3,000. Restoration budgets are 30, 100, 300 and 1,000. The primary random intervention uses eight draws; secondary and restoration controls use three. Three single-feature interventions are also evaluated.

Table 3: Spatial box injection, mean percent closure. Test-frame ranking is a diagnostic; activation-change and raw random controls use the same displayed budgets.

Ranking	10	30	100	300	1,000	3,000
Discovery attribution	0.19	0.40	1.49	4.65	13.61	23.65
Test-frame attribution	4.6	9.6	16.7	23.4	28.5	29.2
Activation change	−0.1	0.0	0.5	1.1	2.6	4.6
Raw random	0.0	0.0	0.0	0.0	0.6	1.2

The full supplementary tables preserve training and validity outputs, all family-specific method results, restoration, token scopes, budget curves, paired contrasts, selectivity replication, first-order diagnostics, in-sample comparisons and every rollout outcome. The file audit also covers the notebook, logs, status records, environment information, binary arrays, checkpoints, figures and all nine example videos. The results archive duplicates extracted files and must not be counted as independent replication. Videos show only task 0 episode 0; the paint video does not display the final pixel-edited policy observation.